Results Interpretation Report

In Silico Network Pharmacology Analysis of Natural Bioactive Compounds (Quercetin, Wogonin, β-Sitosterol, Baicalein, R-Canadine, and Berberine) Targeting Atherosclerosis Mechanisms

Atherosclerosis is a chronic inflammatory condition characterized by the accumulation of plaques within the arterial walls. It is the main driving force behind cardiovascular disease worldwide. Currently, natural bioactive compounds derived from medicinal plants have gained attention as potential therapeutic agents due to their multi-target and multi-pathway pharmacological properties. In this study, network pharmacology analysis was performed to identify potential hub genes and investigate the molecular mechanisms of six natural bioactive compounds, namely Quercetin, Wogonin, β-Sitosterol, Baicalein, R-Canadine, and Berberine, in modulating molecular targets associated with atherosclerosis.

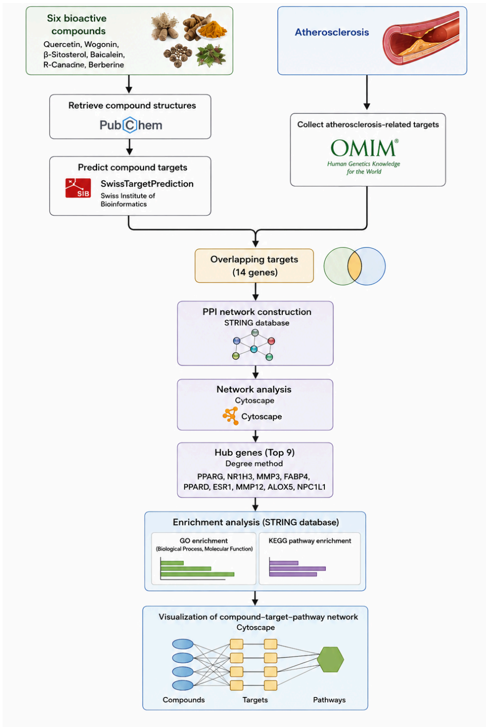

Figure 1. Workflow of the network pharmacology analysis of six bioactive compounds targeting atherosclerosis.

The six bioactive compounds were identified from PubChem, and their potential protein targets were predicted using SwissTargetPrediction, while atherosclerosis-associated genes were retrieved from the OMIM database. The corresponding data are available in Data.xlsx. The predicted compound targets were then intersected with the atherosclerosis-associated genes to identify overlapping targets representing potential molecular targets relevant to the disease. These targets were submitted to STRING-DB to construct a protein–protein interaction (PPI) network using a confidence score of ≥0.400, followed by network visualization and topological analysis in Cytoscape. Gene Ontology (GO) and KEGG pathway enrichment analyses were performed on the overlapping targets using STRING-DB. Finally, the compound–target, and enriched pathway networks were integrated and visualized in Cytoscape.

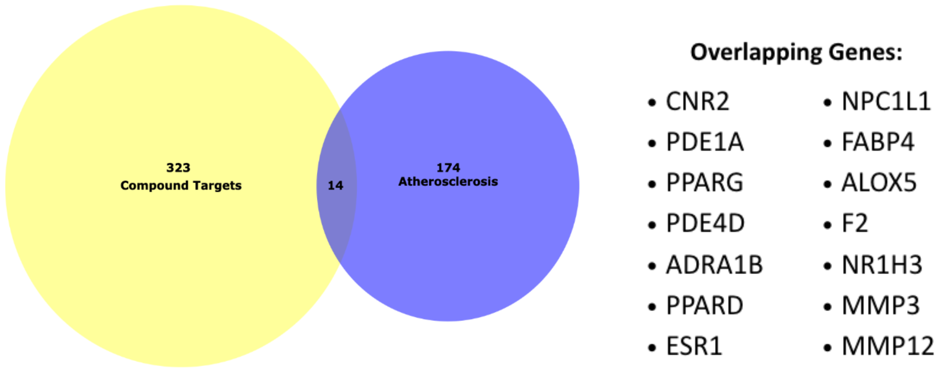

Figure 2. Intersection of compound-predicted targets and atherosclerosis-associated genes.

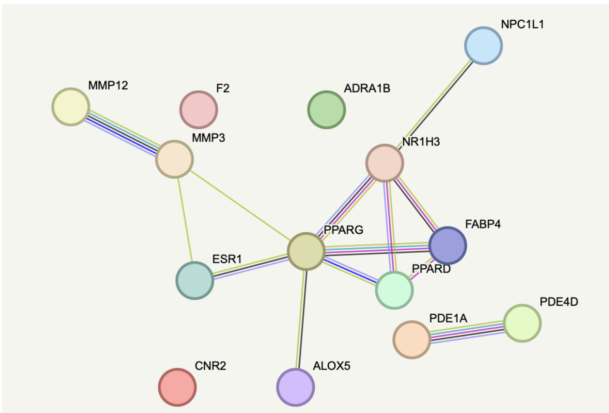

Figure 3. Protein–Protein Interaction (PPI) network of the 14 overlapping genes constructed using STRING.

Analysis of the target intersection identified 14 overlapping genes between atherosclerosis-associated genes and the predicted targets of the six bioactive compounds (Figure 2). These overlapping genes were considered potential molecular targets through which the selected compounds may modulate atherosclerosis-related mechanisms. The 14 overlapping genes were subsequently submitted to STRING to construct a protein–protein interaction (PPI) network, which was visualized and analyzed in Cytoscape using the cytoHubba plugin. Based on the Degree method, nine hub proteins were identified (Table 1), with PPARG showing the highest connectivity (Degree = 6), followed by NR1H3 (Degree = 4). PPARG (peroxisome proliferator-activated receptor gamma) and NR1H3 (nuclear receptor subfamily 1 group H member 3) also showed the highest betweenness and closeness centrality values among the analyzed proteins (Table 2), indicating their relatively central positions within the PPI network. These findings suggest that PPARG and NR1H3 may represent key molecular targets involved in the interactions among the identified atherosclerosis-related proteins.

Table 1. Top 9 Hub Proteins Ranked by Degree Centrality from Overlapping Genes..

Rank	Name	Score
1	PPARG	6.0
2	NR1H3	4.0
3	MMP3	3.0
3	FABP4	3.0
3	PPARD	3.0
6	ESR1	2.0
7	MMP12	1.0
7	ALOX5	1.0
7	NPC1L1	1.0

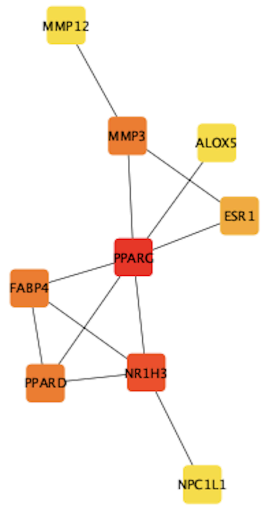

Table 2. Topological parameters of selected hub proteins in the PPI network.

Name	Degree	Betweenness Centrality	Closeness Centrality
FABP4	3	00.00	0.7142857142857143
ALOX5	1	00.00	00.05
NR1H3	4	00.04	0.8333333333333334
PPARG	4	00.04	0.8333333333333334
PPARD	3	00.00	0.7142857142857143
NPC1L1	1	00.00	00.05

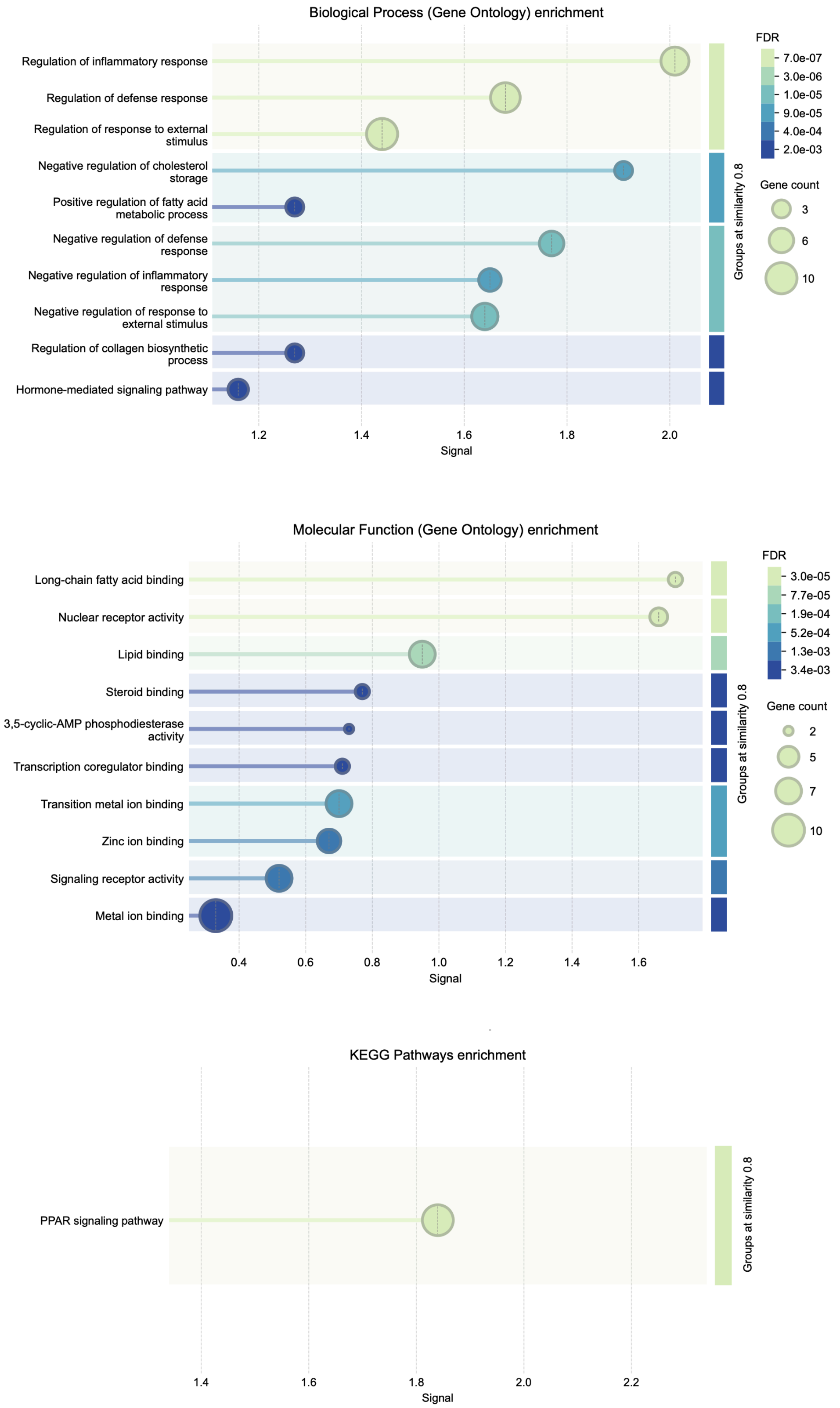

Figure 4. Gene Ontology (GO) enrichment analysis of Biological Process, Molecular Function, and KEGG pathway enrichment analysis of the overlapping targets.

The Biological Process enrichment analysis revealed significant involvement in the regulation of inflammatory response and lipid metabolic processes, two major biological mechanisms underlying atherosclerosis. Besides, molecular function analysis showed that long-chain fatty acid binding was significantly enriched, which is particularly relevant to FABP4, one of the identified hub proteins and a protein involved in fatty acid handling. In addition, the KEGG analysis identified the PPAR signaling pathway, which is directly associated with PPARG, one of the most highly connected hub proteins in the PPI network. Together, these findings suggest that the overlapping targets may influence atherosclerosis through interconnected mechanisms involving inflammation, lipid metabolism, fatty acid regulation, and PPAR-mediated signaling.

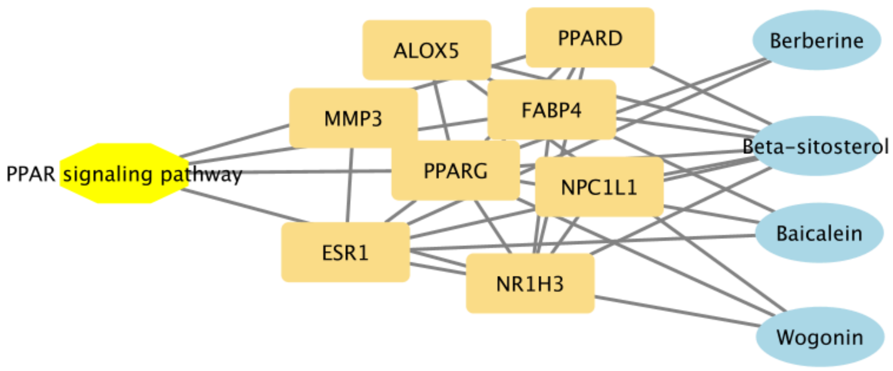

Figure 5. Integrated compound–target–pathway network of the selected bioactive compounds, overlapping targets, and PPAR signaling pathway.

These findings were further integrated into compound–target–pathway network shows the potential multi-target and multi-pathway interactions of the selected bioactive compounds (Figure 5). Several compounds were connected to multiple overlapping targets, while some targets were shared among different compounds, suggesting potential convergent effects on atherosclerosis-related molecular mechanisms. Overall, the network highlights the potential involvement of the selected compounds in modulating PPAR-mediated mechanisms associated with atherosclerosis.